

Design Proposal: Mach–Zehnder Interferometer

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Abstract: This report proposes a set of unbalanced Mach–Zehnder interferometers (MZIs) for extracting the group index of a silicon strip waveguide from the measured free spectral range (FSR). The baseline design uses a 220 nm-high, 500 nm-wide strip waveguide operated in the fundamental quasi-TE mode, with Y-branches as splitter and combiner. Waveguide dispersion will be calculated with an eigenmode solver and represented by a polynomial compact model. Six unbalanced MZIs plus one balanced control are proposed. Their simulated spectra will later be compared with fabricated-device measurements to experimentally determine the group index.

1. Introduction

Silicon photonics enables dense integration of passive and active optical components using a fabrication platform compatible with large-scale semiconductor processing [1, 2]. Among the most useful passive building blocks is the Mach–Zehnder interferometer (MZI), in which an input field is divided into two optical paths and recombined. A difference in optical path length produces a wavelength-dependent phase difference and therefore a periodic transmission spectrum.

The objective of this project is not to optimize an active modulator, but to use an unbalanced MZI as a metrology structure for measuring the group index of a silicon waveguide. The free spectral range (FSR) of the MZI is set primarily by the waveguide group index and the path-length imbalance. Consequently, a measured transmission spectrum can be used to extract the experimental group index and directly compare it with the value predicted by waveguide simulation.

The proposed baseline waveguide is a 220 nm-high and 500 nm-wide silicon strip waveguide, operated in the fundamental quasi-TE mode near 1550 nm. The circuit uses the course-provided TE grating couplers and Y-branches, while the principal design variable is the arm-length difference

$$\Delta L = L_2 - L_1. \quad (1)$$

A set of different values of ΔL is included so that the expected inverse relation between FSR and path imbalance can be tested experimentally.

2. Theory

2.1. Waveguide effective index and group index

For a guided mode with propagation constant β , the modal effective index is

$$n_{\text{eff}}(\lambda) = \frac{\beta(\lambda)\lambda}{2\pi}. \quad (2)$$

Because both the material indices and modal confinement vary with wavelength, n_{eff} is dispersive. The group index is [2]

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}(\lambda)}{d\lambda}. \quad (3)$$

The group velocity is $v_g = c/n_g$. In this work, $n_{\text{eff}}(\lambda)$ and $n_g(\lambda)$ will be obtained from a wavelength sweep of the fundamental quasi-TE mode using the finite-difference eigenmode (FDE) solver in Ansys Lumerical MODE [3, 4].

2.2. Mach–Zehnder transfer function

Consider an ideal MZI with a 50:50 splitter and a 50:50 combiner. The two arms have lengths L_1 and L_2 and share the same waveguide cross section. If the power-attenuation coefficient is α and the propagation constant is β , the field contributions arriving at one output port can be written as

$$E_{\text{out}} = \frac{E_{\text{in}}}{2} \left[e^{-\alpha L_1/2} e^{-j\beta L_1} + e^{-\alpha L_2/2} e^{-j\beta L_2} \right]. \quad (4)$$

The corresponding normalized power transmission is

$$\begin{aligned} T(\lambda) &= \frac{P_{\text{out}}}{P_{\text{in}}} \\ &= \frac{1}{4} \left[e^{-\alpha L_1} + e^{-\alpha L_2} + 2e^{-\alpha(L_1+L_2)/2} \cos(\beta\Delta L + \phi_0) \right], \end{aligned} \quad (5)$$

where ϕ_0 represents any wavelength-independent phase offset introduced by the splitter/combiner convention. If propagation loss is neglected and both couplers are ideal,

$$T(\lambda) = \frac{1}{2} \left[1 + \cos\left(\frac{2\pi n_{\text{eff}}(\lambda)\Delta L}{\lambda} + \phi_0\right) \right]. \quad (6)$$

The phase offset changes the absolute locations of the fringes but not their spacing. Therefore, the FSR remains a robust quantity for extracting the group index.

2.3. Derivation of group index from the FSR

The wavelength-dependent differential phase is

$$\Delta\phi(\lambda) = \beta(\lambda)\Delta L. \quad (7)$$

Two neighboring transmission maxima are separated by one full phase cycle, so

$$2\pi \approx \left| \frac{d(\beta\Delta L)}{d\lambda} \right| \Delta\lambda, \quad (8)$$

where $\Delta\lambda$ is the FSR. Since

$$\frac{d\beta}{d\lambda} = -\frac{2\pi n_g}{\lambda^2}, \quad (9)$$

the wavelength-domain FSR is

$$\boxed{\text{FSR}(\lambda) \approx \frac{\lambda^2}{n_g(\lambda)|\Delta L|}}. \quad (10)$$

Rearranging Eq. (10) gives the experimental group index:

$$\boxed{n_g(\lambda_c) \approx \frac{\lambda_c^2}{|\Delta L| \text{FSR}}}. \quad (11)$$

For two adjacent maxima (or minima) at λ_1 and λ_2 , a convenient finite-spacing form is

$$\boxed{n_g \approx \frac{\lambda_1 \lambda_2}{|\Delta L| |\lambda_2 - \lambda_1|}}. \quad (12)$$

Equation (12) will be used on the measured spectrum. Using many neighboring fringes rather than a single pair will reduce sensitivity to wavelength noise and peak-picking error.

3. Modeling and Simulation

3.1. Waveguide geometry and eigenmode simulation

The baseline waveguide geometry is summarized in Table 1. The 220 nm silicon device-layer thickness is fixed by the fabrication process, and the default 500 nm width is retained so that the design remains compatible with the standard components supplied in the course. Only the fundamental quasi-TE mode is considered in the initial proposal.

Table 1. Baseline waveguide and simulation parameters.

Parameter	Proposed value
Waveguide type	Silicon strip
Waveguide height	220 nm
Waveguide width	500 nm
Polarization	Fundamental quasi-TE
Design wavelength	1550 nm
MODE wavelength sweep	1500 nm to 1600 nm
Splitter/combiner	Course-provided Y-branch
Input/output coupling	Course-provided TE grating couplers

The 2-D cross section will be solved in Lumerical MODE using the FDE solver. A wavelength sweep will track the same quasi-TE mode across the full range. The mode identity will be checked from the field distribution and TE polarization fraction, and mesh convergence will be verified before final values are reported. At the design wavelength, the simulated values will be reported

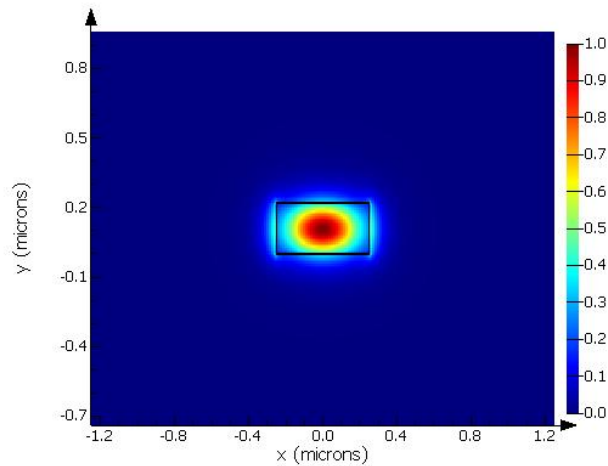


Fig. 1. Simulated electric-field intensity of the fundamental quasi-TE mode of the 220 nm×500 nm strip waveguide.

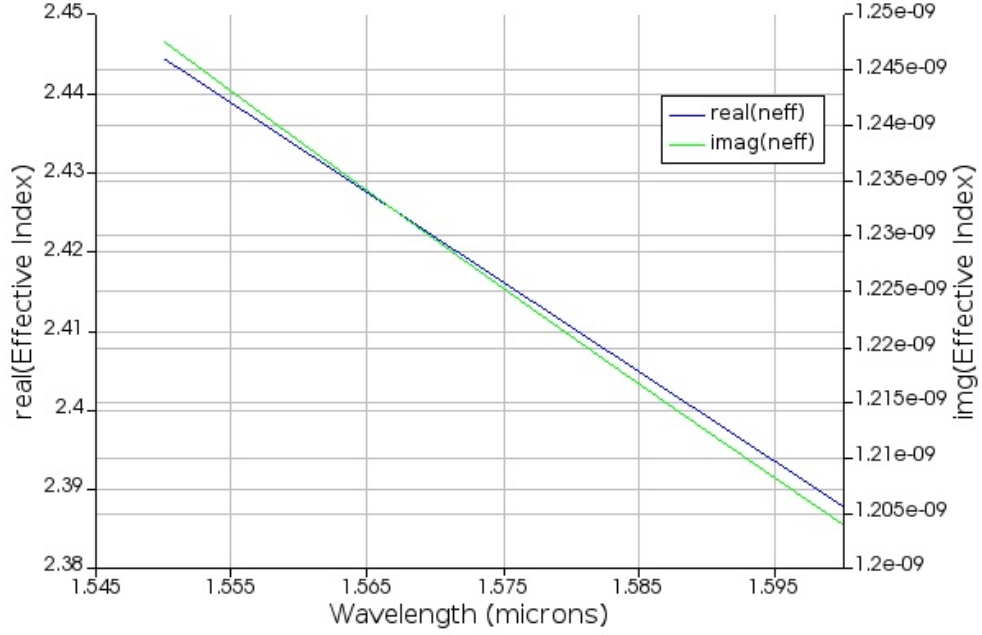


Fig. 2. Simulated effective index n_{eff} versus wavelength for the fundamental quasi-TE mode.

as

$$n_{\text{eff}}(1550 \text{ nm}) = 2.445, \quad (13)$$

$$n_g(1550 \text{ nm}) = 4.196. \quad (14)$$

3.2. Compact model for the waveguide

The simulated effective index will be represented by a polynomial around $\lambda_0 = 1.55 \mu\text{m}$. A second-order model is sufficient as a starting point over the relatively narrow wavelength interval:

$$n_{\text{eff}}(\lambda) = a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2, \quad (15)$$

where λ is expressed in micrometers. The fitted coefficients are

$$a_0 = 2.44733, \quad (16)$$

$$a_1 = -1.13281, \quad (17)$$

$$a_2 = -0.0411206. \quad (18)$$

The compact model provides the derivative analytically:

$$\frac{dn_{\text{eff}}}{d\lambda} = a_1 + 2a_2(\lambda - \lambda_0), \quad (19)$$

which can be substituted into Eq. (3) to calculate $n_g(\lambda)$. The polynomial order will only be increased if the residual error of the quadratic fit is not sufficiently small.

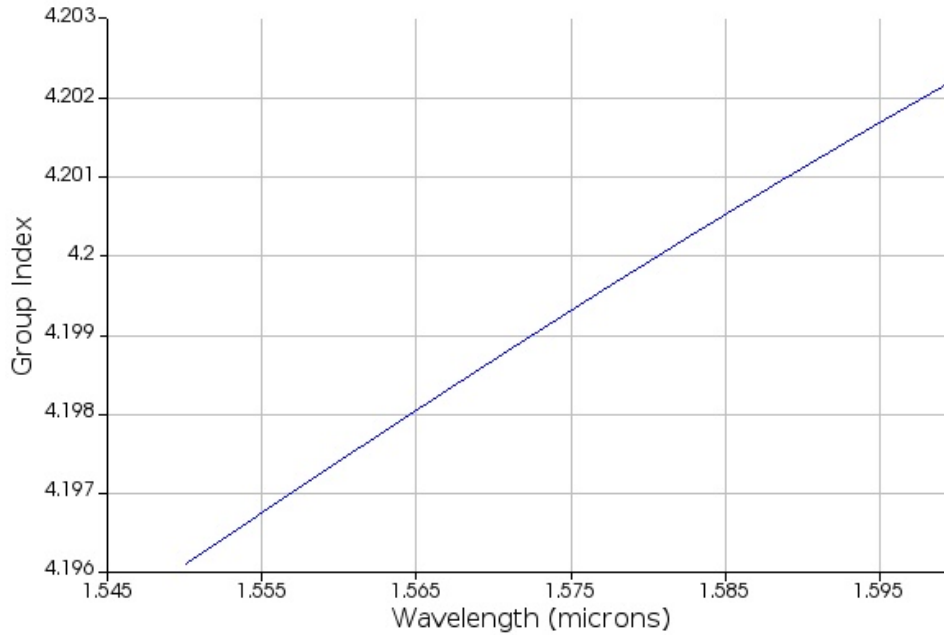


Fig. 3. Simulated group index n_g versus wavelength for the fundamental quasi-TE mode.

3.3. MZI circuit model

The circuit consists of an input grating coupler, a Y-branch splitter, two strip-waveguide arms, a Y-branch combiner, and an output grating coupler. The circuit-level response will be simulated in Ansys Lumerical INTERCONNECT, using course-provided compact/S-parameter models for the couplers and Y-branches where available [5, 6]. Waveguide dispersion will be represented using the MODE-derived compact model.

The first circuit check is the ideal analytical response. Figure 4 shows an example of a circuit layout in Lumerical INTERCONNECT.

3.4. Design variations and expected FSR

The main parameter sweep is the MZI path-length imbalance. Five unbalanced interferometer circuits are proposed. The corresponding FSR is calculated from Eq. (10). All unbalanced designs have an FSR below the approximately 50 nm measurement bandwidth.

Varying only ΔL keeps the waveguide geometry and component interfaces identical among the devices, so the measured trend can be attributed primarily to the interferometer imbalance rather than to a change of waveguide cross section. The balanced device is included as a control, while the range of unbalanced devices provides both coarse and dense spectral fringes.

4. Fabrication and Layout

The layout will be implemented in KLayout using the SiEPIC tools and the process design kit supplied in the course [7, 8]. The layout will use the verified course components for the grating couplers and Y-branches. Routing will preserve the intended ΔL values while respecting the process design rules and minimum bend-radius requirements. The final version of this section will include a labeled layout image, the final footprint, bend radii, component names, and verification results.

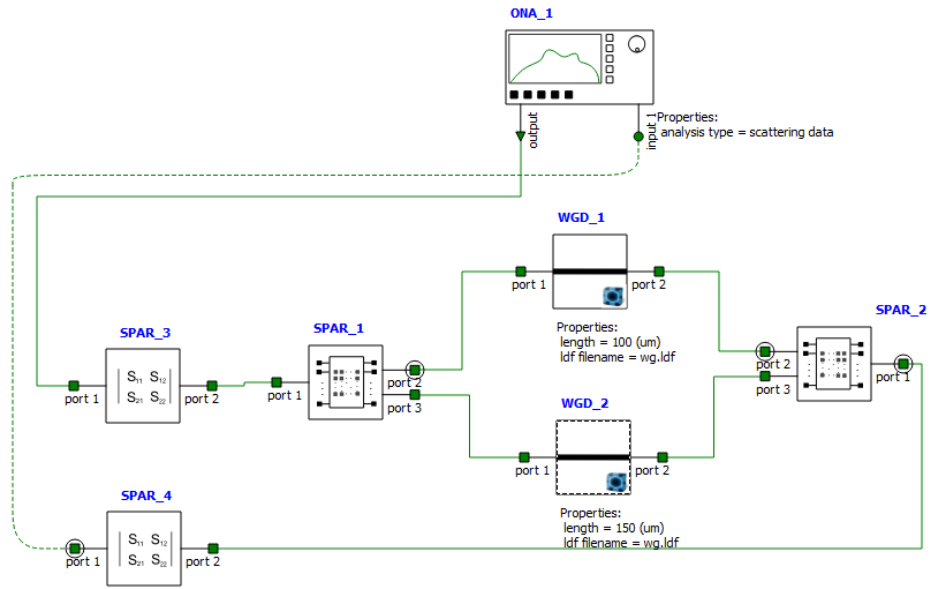


Fig. 4. MZI model implemented in the INTERCONNECT software.

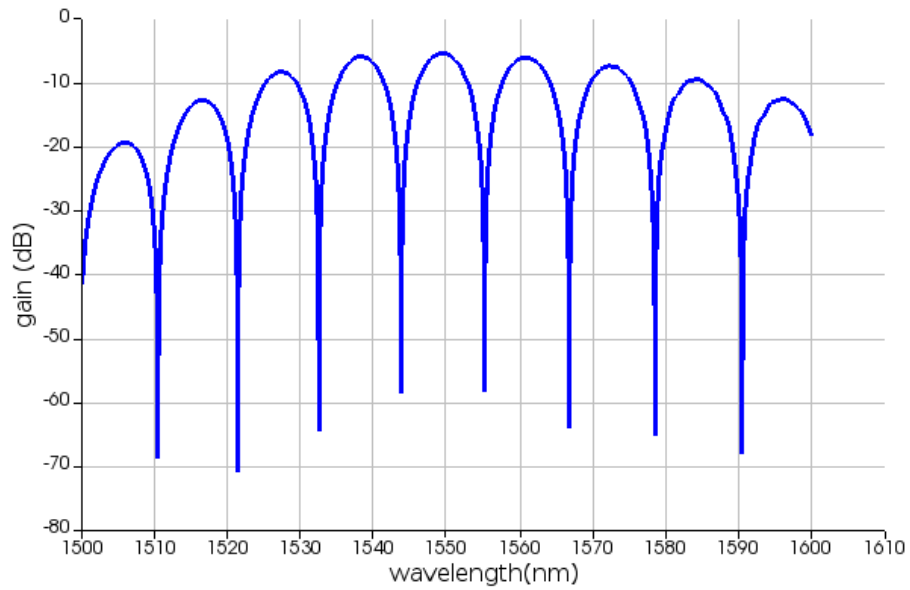


Fig. 5. Transfer function of an MZI with $\Delta L = 50 \mu\text{m}$.

Table 2. Proposed MZI variants. Predicted FSR values use $\lambda_0 = 1550$ nm.

Design	$ \Delta L $ (μm)	FSR (nm)
MZI-50	50	11.64
MZI-75	75	7.5
MZI-100	100	5.72
MZI-125	125	4.57
MZI-150	150	3.81

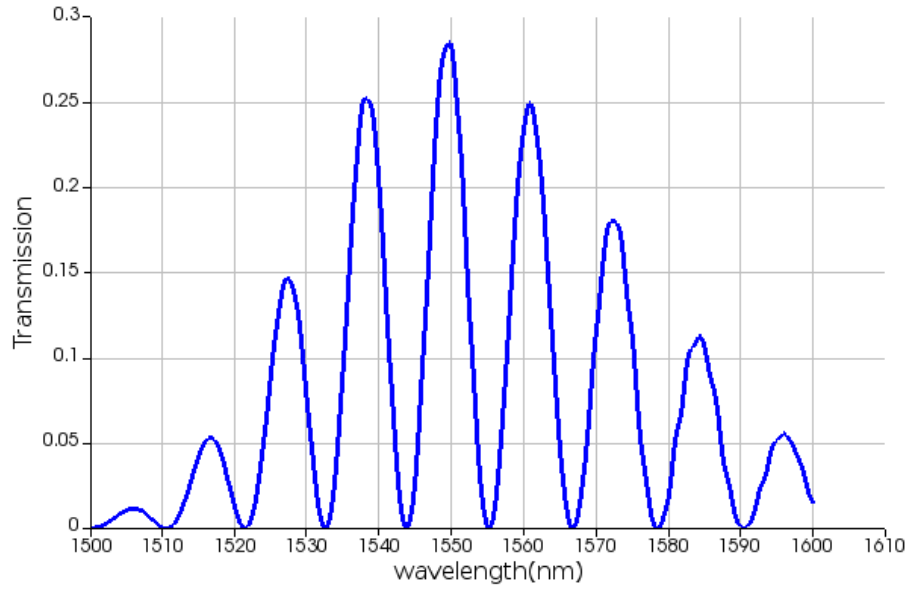


Fig. 6. Transmission of an MZI with $\Delta L = 50 \mu\text{m}$.

5. Experimental Plan

After fabrication, the transmission spectrum of each MZI will be measured over the wavelength range supported by the test setup and grating couplers. For each unbalanced MZI, neighboring transmission maxima or minima will be identified and their wavelength spacing will be recorded. The experimental group index will then be calculated using Eq. (12). To reduce uncertainty, several adjacent FSR values will be extracted from each spectrum and averaged or fit as a function of wavelength.

The measured values will be compared with the MODE prediction $n_g(\lambda)$. Agreement among MZIs with different ΔL will provide an internal consistency check. Any systematic discrepancy will be discussed in terms of fabrication-induced changes in waveguide width/thickness, material dispersion, component response, and wavelength-calibration uncertainty.

6. Experimental Data

7. Analysis

8. Conclusion

A family of unbalanced silicon MZIs is proposed for extracting the group index of a $220\text{ nm}\times 500\text{ nm}$ strip waveguide in the quasi-TE polarization. The design uses a balanced control and six path-length imbalances from $20\text{ }\mu\text{m}$ to $120\text{ }\mu\text{m}$. After fabrication, the measured FSR will be used to obtain the experimental group index and compare it directly with simulation.

References

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